Probability and Statistics using R

Lab Session-3

Summary STatistics (Location, spread, shape and peakedness)

Course Code : MAT445

Date:30.07.2026 Time : 3:15PM - 4:15 PM

Programming Tasks

1. Create a vector of marks: 45, 56, 67, 72, 81, 90, 76, 68 and find the mean.
2. Create a vector of salaries: 25000, 28000, 32000, 30000, 29000, 31000, 27000 and find the median.
3. Find the minimum, maximum and range of the data:

   $$12,\ 15,\ 18,\ 20,\ 24,\ 30,\ 35,\ 40$$

4. Compute the variance and standard deviation for the data:

   $$14,\ 18,\ 20,\ 22,\ 24,\ 28,\ 30$$

5. Find the first quartile, median and third quartile of:

   $$5,\ 8,\ 10,\ 12,\ 15,\ 18,\ 20,\ 22,\ 25$$

6. Find the interquartile range (IQR) for:

   $$16,\ 18,\ 21,\ 23,\ 25,\ 28,\ 30,\ 35,\ 40$$

7. Display the complete summary statistics of:

   $$55,\ 62,\ 70,\ 68,\ 75,\ 80,\ 90,\ 85$$

8. Install the `moments` package and calculate the skewness of the following data:

   $$10,\ 12,\ 15,\ 18,\ 22,\ 40,\ 45$$

9. Calculate the kurtosis of the following observations:

   $$5,\ 7,\ 9,\ 10,\ 12,\ 15,\ 20,\ 25,\ 50$$

10. Write an R program to calculate Mean, Median, Variance, Standard Deviation and IQR of a given dataset using a single program.